

7th Heaven Cafe: Database Design

This document outlines the database design for the 7th Heaven Cafe Web Platform, implemented using Google Cloud Firestore (NoSQL) and Firebase Web SDK. The architecture leverages document-based storage, embedding arrays for point-in-time snapshots (order items) while referencing related entities (users and orders).

1. User (Admin & Customer) Collection

Stores registered customers, administrators, and their authentication credentials.

Field Name	Data Type	Constraints / Description
uid	String	Primary Key. Matches Firebase Auth UID.
name	String	Required. Full name of the user or admin.
email	String	Required, Unique. Used for login and invoices.
phone	String	Optional. Contact phone number.
role	String	Required. Enum: ['customer', 'kitchen', 'admin']. Default: 'customer'.
addresses	Array of Objects	Saved customer delivery address list.
createdAt	Timestamp	Timestamp when the account was registered.

2. MenuItem Collection

Stores individual food and beverage items offered by 7th Heaven Cafe.

Field Name	Data Type	Constraints / Description
id	String	Primary Key, Auto-generated document ID.
name	String	Required. Name of the food/beverage item.
description	String	Optional. Detailed description of the item.
price	Number	Required. Current selling price in INR (₹).
category	String	Required. Category grouping on customer menu.
imageUrl	String	Optional. Cloud Storage URL for item photo.
available	Boolean	Default: true. Toggles customer menu visibility.
addons	Array of Objects	Optional addon customization options with extra prices.
popularity	Number	Default: 0. Tracks total quantity ordered across sales.
createdAt	Timestamp	Timestamp when the menu item was created.

3. Order Collection

Records customer transactions, encapsulating a point-in-time snapshot of the cart and order lifecycle status.

Field Name	Data Type	Constraints / Description
id	String	Primary Key, Auto-generated document ID.
orderNumber	String	Required. Formatted sequential order ID (e.g., '7H-20260303-0042').
userId	String	Required. Reference to User uid, or 'guest'.
customerName	String	Required. Name provided by customer.
customerEmail	String	Required. Email address for invoice receipt.
customerPhone	String	Required. Phone number for delivery updates.
customerAddress	String	Required. Complete delivery address.
items	Array of Objects	Required. Embedded historical snapshot of items.
items.menuItemId	String	Required. Reference to MenuItem document ID.
items.name	String	Required. Snapshot of item name at time of order.
items.price	Number	Required. Snapshot of item price at time of order.
items.quantity	Number	Required. Number of units ordered.
subtotal	Number	Sum of (price * quantity) for all line items.
taxes	Number	Calculated 5% GST tax amount.
deliveryFee	Number	Delivery charge (₹40 or ₹0 if subtotal ≥ ₹500).
totalAmount	Number	Required. Final net payable amount.
paymentMethod	String	Required. Enum: ['Online (Razorpay)', 'Cash on Delivery'].
status	String	Required. Enum: ['Placed', 'Preparing', 'Out for Delivery', 'Delivered']. Default: 'Placed'.
paymentStatus	String	Required. Enum: ['Pending', 'Paid', 'Failed'].
createdAt	Timestamp	Timestamp of when the order was placed.
estimatedDeliveryAt	Timestamp	Calculated delivery ETA for live customer tracking.

4. Counter Collection

A utility collection used to generate concurrent-safe, sequential order numbers for customer invoices and kitchen ticketing.

Field Name	Data Type	Constraints / Description
_id	String	Primary Key (e.g., 'order_counter').
seq	Number	Required. Tracks the last assigned order sequence number. Default: 0.
updatedAt	Timestamp	Timestamp of last sequence increment.

Relationships & Data Integrity Rules

- Multi-Role Isolation:** Every User, MenuItem, and Order document is protected by Firestore Security Rules. Customers have strictly scoped access to read/write their own orders and profile documents, while kitchen staff and administrators possess elevated privileges for status progression and menu catalog updates.
- Snapshot Pattern (OrderItems):** Items within an order are embedded as an array of objects rather than traditional normalized foreign keys. The item's name and price are captured at the exact time of purchase. This guarantees that if an admin later updates the price of a MenuItem or deletes it entirely, historical order totals remain accurate and immutable.
- Auto-Incrementing Sequences:** The Counter collection allows atomic increment operations via Firestore transactions to generate gapless sequential order numbers, avoiding race conditions during concurrent customer order placements.